

Lesson 2 — Build the Model & Sequence It

Matches LEGO Teacher Portal lesson: "Get the Gold".

Objective (I can...): I can build the unit model and put blocks in the right order so it does steps one after another. **CS concept:** Sequencing (instructions run top to bottom, in order) · **Builds on:** Lesson 1 (running a program).

AI Connection (2 min): A voice assistant answering follows an ordered recipe: hear → understand → reply. *First-principles question:* "What would happen if AI did those steps out of order?"

Materials: LEGO CS & AI kit (1 per pair), device per pair, 4 sequence picture cards per pair. **This is the unit's build lesson** — build the Basics model now and keep it assembled through Lesson 5.

60-minute flow - Warm-up (8): Card scramble — pairs put 4 out-of-order "make toast" cards in the right order and discuss how wrong order breaks the result. - **Concept + demo (10):** Show two programs with the same blocks in different orders; run both so students see different outcomes. Name the top-to-bottom rule. - **Build/Code (30):** Pairs **build the Basics model** per the Canvas instructions (this is the one build for the unit), then write a 4-step sequence (light on → motor spins → wait → motor stops) and run it, checking each step in order. *Swap at 15 min.* - **Challenge (7):** Rearrange the same blocks into a new sensible behavior and describe what changed. - **Share + pack-up (5):** Each pair shows original vs. rearranged program. **Leave the model built** — store it on its tray.

Pair roles: During the build, Driver connects bricks while Navigator reads steps; during coding they swap. *Swap at 15 min. (Trio rotates build/read/test.)*

Differentiation: *Support* — a worked "twin" 4-step example on paper to adapt (not copy). *Extension / fast-finisher* — build a 6-step sequence. *Reteach* — number the blocks 1-4 out loud together and run one step at a time.

Exit check: Pair produces a working ordered sequence and can explain why order changed the result.

At-home (optional): Draw a 4-step comic of waking up, in order. ~10 min.
